

Predicting Healthy Life Expectancy

Using Machine Learning with Environmental,
Nutritional, and Health System Indicators



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Problem & Motivation

7.4 vs 6.3

years gained in life expectancy vs HALE
(GBD 2017 — 1.1 years spent in poor health)



Gap in prediction targets

Most ML studies predict general life expectancy, not HALE (years in good health)



Narrow feature sets

Models use only socio-economic and clinical indicators — missing environmental, nutritional, and health-system factors



Weak validation

Prior work uses random splits, which leak future data into training sets

Our Approach

Extend Pranjal et al. [2025] HALE prediction framework by adding 3 upstream health variables from the World Bank, and adopt a temporal train-test split for realistic evaluation.



PM2.5
Air Pollution



Prevalence of
Undernourishment



Health
Expenditure

Related Work & Our Novelty

Study	Target	Best R ²	Env.	Nutr.	Health Sys.
Kumari & Siddiqui [2]	Life Exp.	0.96	✗	✗	✗
Sharma et al. [3]	Life Exp.	0.97	✗	✗	✗
Lipesa et al. [8]	Life Exp.	RMSE 2.4	✗	✗	✗
Pranjal et al. [7]	HALE_60	0.99	✗	✗	✗
This Study	HALE_60	0.98	✓	✓	✓

What We Add

1. Target HALE_60

— not general life expectancy

2. Three upstream variables

— PM2.5, undernourishment, health expenditure (from World Bank)

3. Temporal validation

— train 2000–2014, test 2015–2019 (replaces random split)

Methodology

WHO GHO
(8 indicators)

World Bank WDI
(3 new indicators)

Merge → Preprocess → Scale

Backward fill · Interpolation · Mean fallback
Z-score (rates) · MinMax (percentages)

Baseline
8 features + engineered

VS

Extended
11 features + engineered

3 Models: Linear Regression · Random Forest · Gradient Boosting

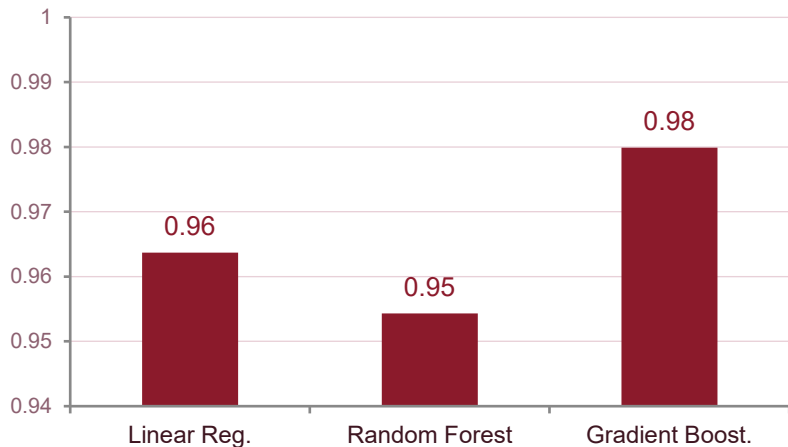
Metrics: RMSE · R^2 · MAE · CV-RMSE

Key Details

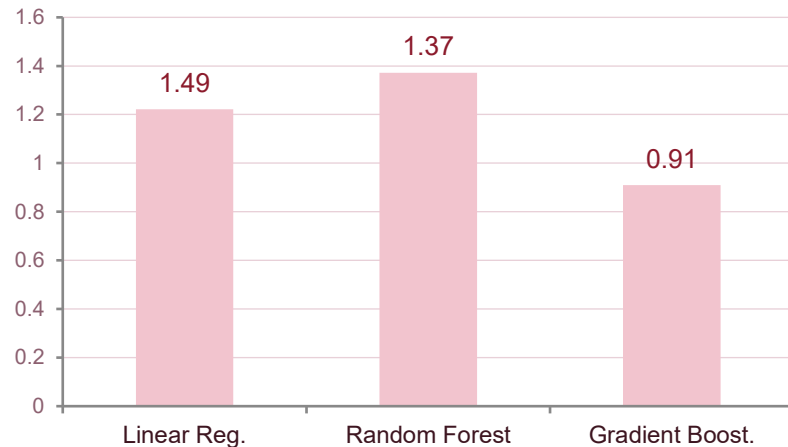
- ✓ 185 countries
- ✓ 2000–2019 (20 years)
- ✓ 3,700 observations
- ✓ Target: HALE_60
- ✓ Temporal split:
Train 2000–2014
Test 2015–2019
- ✓ Scaling after split
(no data leakage)

Results: Extended Feature Set

Test R^2



Test RMSE (years)

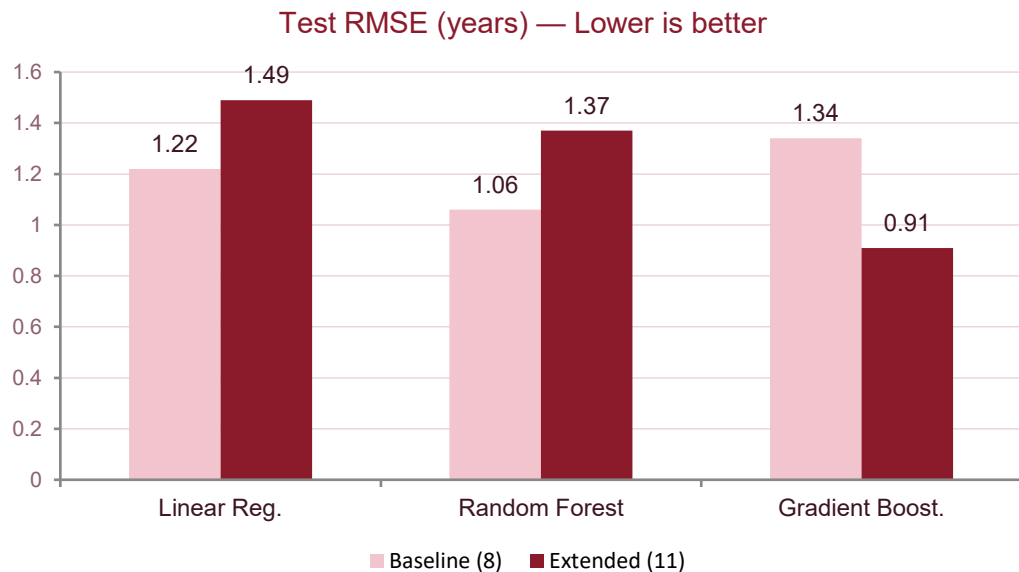


Best Model: Gradient Boosting

$R^2 = 0.9799$ · RMSE = 0.91 years · MAE = 0.61 years

PM2.5 ranked among top features — supporting environmental indicators in HALE prediction

Ablation Study: 8 vs 11 Features



Key Insight

Gradient Boosting is the only model to improve RMSE with extended features (1.34 → 0.91 years).

Linear Regression and Random Forest show slight R^2 decrease — likely due to multicollinearity and feature dilution in simpler models.

→ *Upstream determinants carry modest but genuine signal, best captured by boosting methods.*

Temporal split (2000–2014 train / 2015–2019 test) explains lower absolute metrics vs. original paper's random split.

Conclusion & Limitations



Conclusions



Gradient Boosting achieves $R^2 = 0.98$ and RMSE = 0.91 years on HALE_60



PM2.5 contributes meaningfully to model predictions



Upstream variables help boosting methods but not simpler models — model choice matters



Temporal validation provides stricter, more realistic evaluation

Limitations

- HALE_Birth with/without comparison pending
- Country-level data misses within-country variation
- PM2.5 data sparse in early years (interpolated)

Future Work

- Expand to 7 ML models
- HALE_Birth sensitivity analysis
- Explore deep learning approaches

Thank You

Questions?

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GitHub: github.com/HALA-Prediction-466/HALE-prediction